Ex. The variables X and Y are connected by the equation $ax + by + c = 0$. Show that the correlation between them is -1 if the signs of a and b are alike and $+1$ if they are different.

$$ax + by + c = 0$$
$$aE(X) + bE(Y) + c = 0$$

$\therefore$ $a[X - E(X)] + b[Y - E(Y)] = 0$

$\Rightarrow$ $[X - E(X)] = -\frac{b}{a}[Y - E(Y)]$

$\mathrm{Cov}(X,Y) = E[(X - E(X))(Y - E(Y))] = -\frac{b}{a} E[\{Y - E(Y)\}^2] = -\frac{b}{a}\sigma_y^2$

and $\sigma_x^2 = E[X - E(X)]^2 = \frac{b^2}{a^2}\cdot\sigma_y^2$

$$\gamma_{xy} = \frac{\mathrm{Cov}(X,Y)}{\sigma_x \cdot \sigma_y} = \frac{-\frac{b}{a}\sigma_y^2}{\sqrt{\frac{b^2}{a^2}\cdot\sigma_y^2}\cdot\sqrt{\sigma_y^2}} = \frac{-\frac{b}{a}\sigma_y^2}{\left|\frac{b}{a}\right|\sigma_y^2}$$

$$= \frac{\mp\frac{b}{a}}{\left|\frac{b}{a}\right|} = \begin{cases} +1 & \text{if } a \,\&\, b \text{ have same sign} \\ -1 & \text{if } a \,\&\, b \text{ have opposite sign} \end{cases}$$

Ex. If $Z = ax + by$ and r is the correlation coefficient between X and Y, (a) Show that $\sigma_Z^2 = a^2\sigma_X^2 + b^2\sigma_y^2 + 2ab\, r\, \sigma_X \sigma_Z$

(b) Show that

$$\gamma_{xy} = \frac{(\sigma_x^2 + \sigma_y^2 - \sigma_{x-y}^2)}{2\sigma_x \cdot \sigma_y}$$

$\sigma_x, \sigma_y, \sigma_{x-y}$ are S.D. of X, Y & $X-Y$.

(a) $E(Z) = aE(X) + bE(Y)$

$$Z - E(Z) = a[X - E(X)] + b[Y - E(Y)]$$

$$E(Z - E(Z))^2 = a^2E(X - E(X))^2 + b^2E(Y - E(Y))^2 + 2abE[(X - E(X))(Y - E(Y))]$$

$$\Rightarrow \sigma_Z^2 = a^2\sigma_X^2 + b^2\sigma_y^2 + 2ab\,\mathrm{Cov}(X,Y)$$
$$= a^2\sigma_x^2 + b^2\sigma_y^2 + 2ab\,\gamma_{xy}\sigma_x\sigma_y$$

(b) Taking $a = 1,\ b = -1$

$Z = X - Y$

We get $\sigma_{X-Y}^2 = \sigma_X^2 + \sigma_Y^2 - 2\gamma\sigma_X\sigma_Y$

$$\gamma = \frac{\sigma_X^2 + \sigma_Y^2 - \sigma_{X-Y}^2}{2\sigma_X \cdot \sigma_Y}.$$

Ex. X and Y are two random variables with variances σ_X^2 and σ_Y^2 respectively and γ is the correlation coefficient between them. If $U = X + KY$ and $V = X + \left(\frac{\sigma_X}{\sigma_Y}\right)Y$, find the value of K so that U and V are uncorrelated.

$U = X + KY \qquad E(U) = E(X) + KE(Y)$

$V = X + \frac{\sigma_X}{\sigma_Y}Y \qquad E(V) = E(X) + \frac{\sigma_X}{\sigma_Y}E(Y)$

$U - E(U) = (X - E(X)) + K(Y - E(Y))$

$V - E(V) = (X - E(X)) + \frac{\sigma_X}{\sigma_Y}(Y - E(Y))$

$$\begin{aligned}
Cov(U,V) &= E[(U - E(U))(V - E(V))] \\
&= E\left[\{(X - E(X)) + K(Y - E(Y))\}\{(X - E(X)) + \frac{\sigma_X}{\sigma_Y}(Y - E(Y))\}\right] \\
&= \underline{\sigma_X^2} + \frac{\sigma_X}{\sigma_Y}cov(X,Y) + K\,Cov(X,Y) + \underline{K \cdot \frac{\sigma_X}{\sigma_Y}\sigma_Y^2} \\
&= (\sigma_X^2 + K\sigma_X\sigma_Y) + \left(\frac{\sigma_X}{\sigma_Y} + K\right)Cov(X,Y) \\
&= \sigma_X(\sigma_X + K\sigma_Y) + \left(\frac{\sigma_X + K\sigma_Y}{\sigma_Y}\right)cov(X,Y) \\
&= (\sigma_X + K\sigma_Y)\left[\sigma_X + \frac{Cov(X,Y)}{\sigma_Y}\right] \\
&= (\sigma_X + K\sigma_Y)\left[\sigma_X + \frac{\gamma_{XY}\sigma_X\cancel{\sigma_Y}}{\cancel{\sigma_Y}}\right] = (\sigma_X + K\sigma_Y)(1 + \gamma_{XY})\sigma_X
\end{aligned}$$

If U & V are uncorrelated $Cov(U,V) = 0 \Rightarrow \gamma_{UV} = 0$

$\Rightarrow (\sigma_K + K\sigma_Y)(1 + \gamma_{XY})\sigma_X = 0$

$\Rightarrow \sigma_X + K\sigma_Y = 0 \Rightarrow \boxed{K = -\frac{\sigma_X}{\sigma_Y}}$

Ex. X + Y are indep r.vs

$f(x) = 4ax, \ 0 \le x \le r \qquad f(y) = 4by, \ 0 \le y \le s$

Show that $\text{Corr}(U,V) = \frac{b-a}{b+a}, \quad U = X+Y, \ V = X-Y$

Sol'n of a + b $\quad \int_0^r f(x)dx = 4a\int_0^r x\,dx = 1 \Rightarrow a = \frac{1}{2r^2}$

$\int_0^s f(y)dy = 4b\int_0^s y\,dy = 1 \Rightarrow b = \frac{1}{2s^2}$

$\Rightarrow f(x) = \frac{2x}{r^2}, \ 0 \le x \le r \qquad f(y) = 4by = \frac{2y}{s^2}, \ 0 \le y \le s$

X + Y are indep $\Rightarrow r_{xy} = 0 \Rightarrow \text{Cov}(X,Y) = 0$

$\text{Cov}(U,V) = \text{Cov}(X+Y, X-Y) = \text{Cov}(X,X) - \text{Cov}(XY) + \text{Cov}(YX) - \text{Cov}(YY)$

$= \sigma_X^2 - \sigma_Y^2$

$V(U) = V(X+Y) = V(X) + V(Y) + 2\text{Cov}(X,Y)$

$= \sigma_X^2 + \sigma_Y^2$

$V(V) = V(X-Y) = V(X) + V(Y) - 2\text{Cov}(X,Y)$

$= \sigma_X^2 + \sigma_Y^2$

$r_{UV} = \frac{\text{Cov}(U,V)}{\sigma_U \cdot \sigma_V} = \frac{\sigma_X^2 - \sigma_Y^2}{\sigma_X^2 + \sigma_Y^2}$

$E(X) = \int_0^r x f(x)dx = \frac{2r}{3}, \quad E(X^2) = \int_0^r x^2 f(x)dx = \frac{r^2}{2}$

$V(X) = E(X^2) - (E(X))^2 = \frac{r^2}{18} = \frac{1}{36a}, \quad a = \frac{1}{2r^2}$

Similarly $V(X) = \frac{s^2}{18} = \frac{1}{36b}, \quad b = \frac{1}{2s^2}$

$\therefore r(U,V) = \frac{\left(\frac{1}{36a}\right) - \left(\frac{1}{36b}\right)}{\left(\frac{1}{36a}\right) + \left(\frac{1}{36b}\right)} = \frac{b-a}{b+a}.$

Eq^n of plane of regression

$$\frac{X_1}{\sigma_1} w_{11} + \frac{X_2}{\sigma_2} w_{12} + \frac{X_3}{\sigma_3} w_{13} = 0$$

Where $X_i = x_i - \bar{x}_i$

$$w = \begin{vmatrix} 1 & r_{12} & r_{13} \\ r_{21} & 1 & r_{23} \\ r_{31} & r_{32} & 1 \end{vmatrix}$$

w_{ij} is the cofactor in ith row & jth col^n

Find regression eq^n of x_1 on x_2 and x_3 given the following results.

Trait	Mean	S.D
x_1	28.02	4.42
x_2	4.91	1.10
x_3	594	85

$r_{12} = +0.80$

$r_{23} = -0.56$

$r_{31} = -0.40$

$$(x_1 - \bar{x}_1)\frac{w_{11}}{\sigma_1} + (x_2 - \bar{x}_2)\frac{w_{12}}{\sigma_2} + (x_3 - \bar{x}_3)\frac{w_{13}}{\sigma_3} = 0$$

$$w = \begin{vmatrix} 1 & r_{12} & r_{13} \\ r_{21} & 1 & r_{23} \\ r_{31} & r_{32} & 1 \end{vmatrix} \qquad w_1 = \begin{vmatrix} 1 & r_{23} \\ r_3 & 1 \end{vmatrix}$$

$$= 1 - r_{23}^2 = 1 - (-0.56)^2 = 0.686$$

Reg eq^n

$$\frac{0.686}{4.42}(x_1 - 28.02) + \frac{(-0.576)}{1.10}(x_2 - 4.91) + \frac{(-0.048)}{85.0}(x_3 - 594) = 0.$$

Ans.

$$w_{12} = -\begin{vmatrix} r_{21} & r_{23} \\ r_{31} & 1 \end{vmatrix}$$

$$= -(r_{21} - r_{31} r_{23})$$

$$= r_{31} r_{23} - r_{21}$$

$$= -0.576$$

$$w_{13} = \begin{vmatrix} r_{21} & 1 \\ r_{31} & r_{32} \end{vmatrix}$$

$$r_{21} r_{32} - r_{31} = -0.048$$